



D Data
S Structures
A Algorithms



Data Structures & Algorithms

Today's Topic: Binary Search Tree



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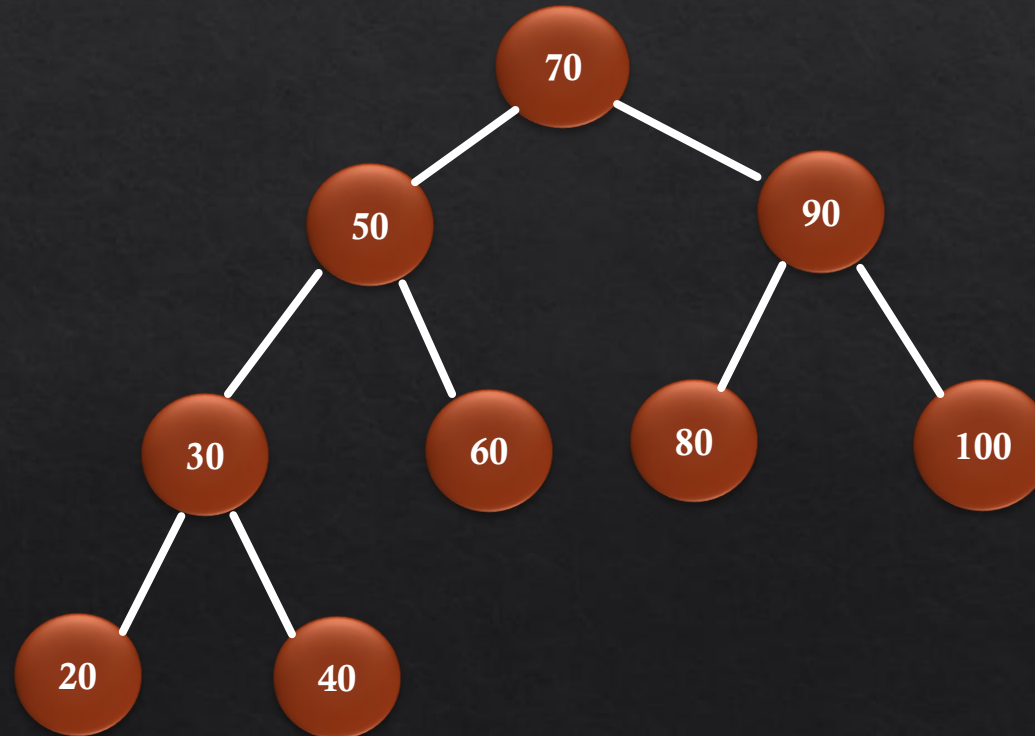


<https://sithuaung-ink.github.io/data-analyst-portfolio/>

4/2/2026

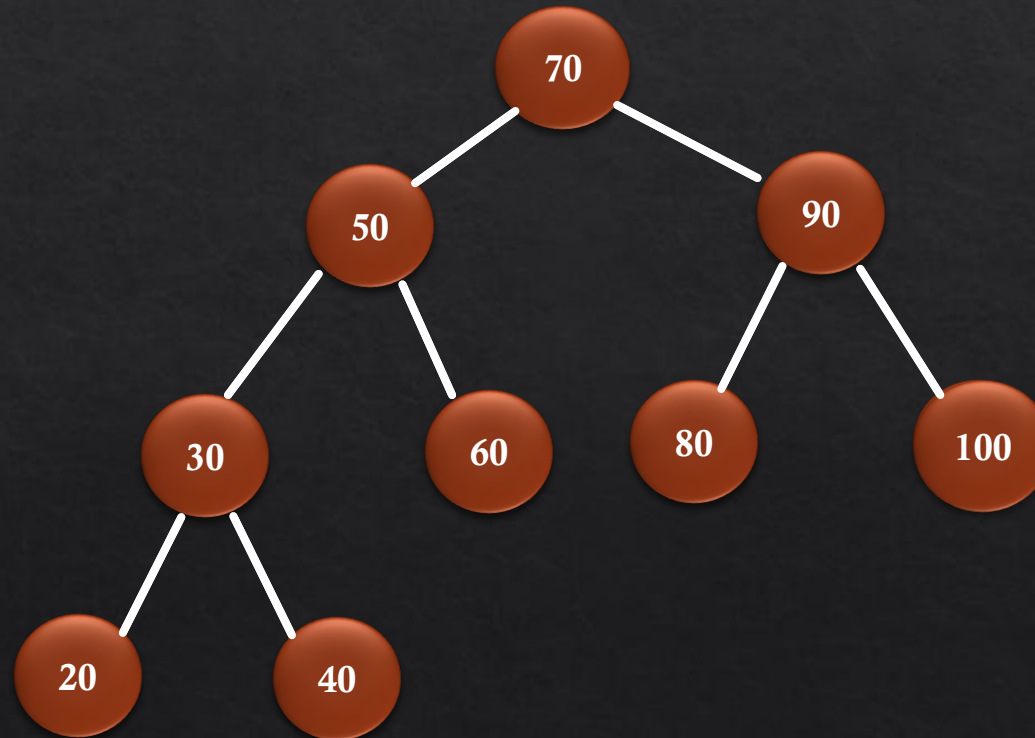
What is a Binary Search Tree (BST)?

- In the left subtree, the value of a node is less than or equal to its parent node's value.
- In the right subtree, the value of a node is greater than its parent node's value.



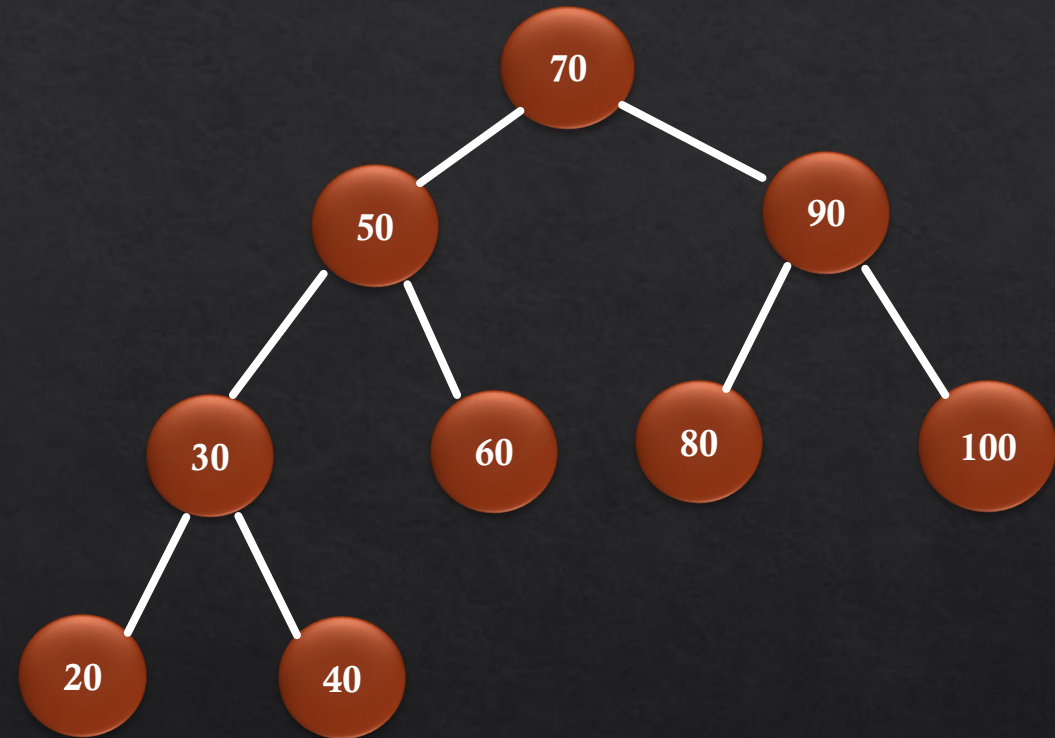
Why does a BST need it?

- It performs faster than a binary tree when inserting and deleting nodes.

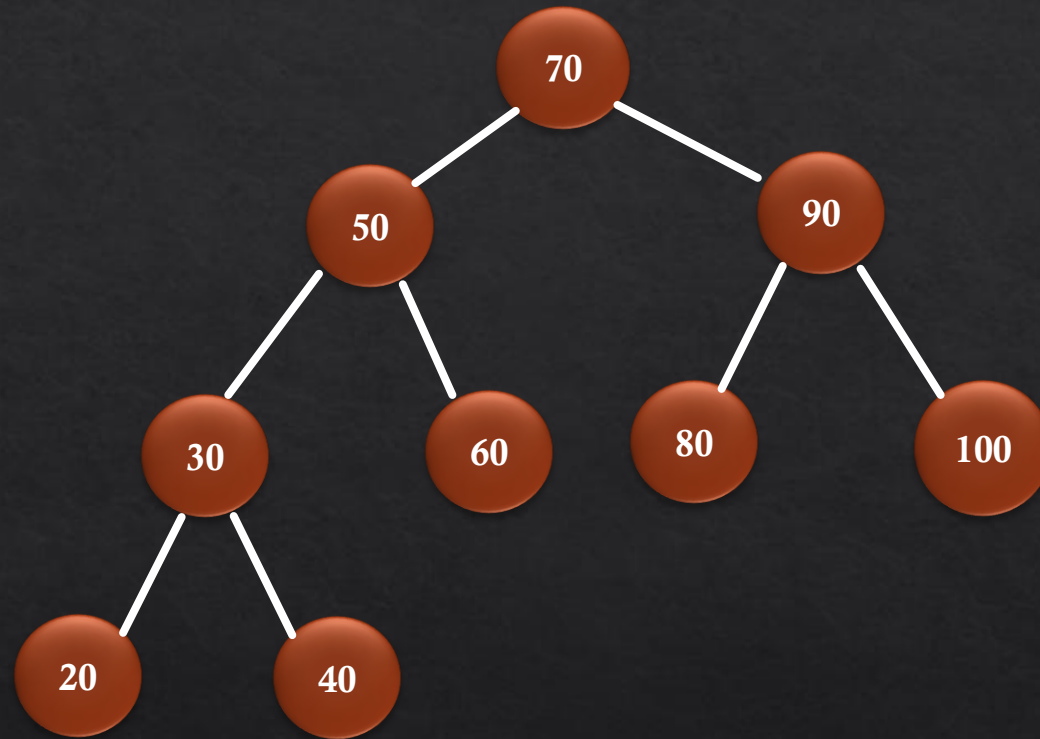


Operations of BST

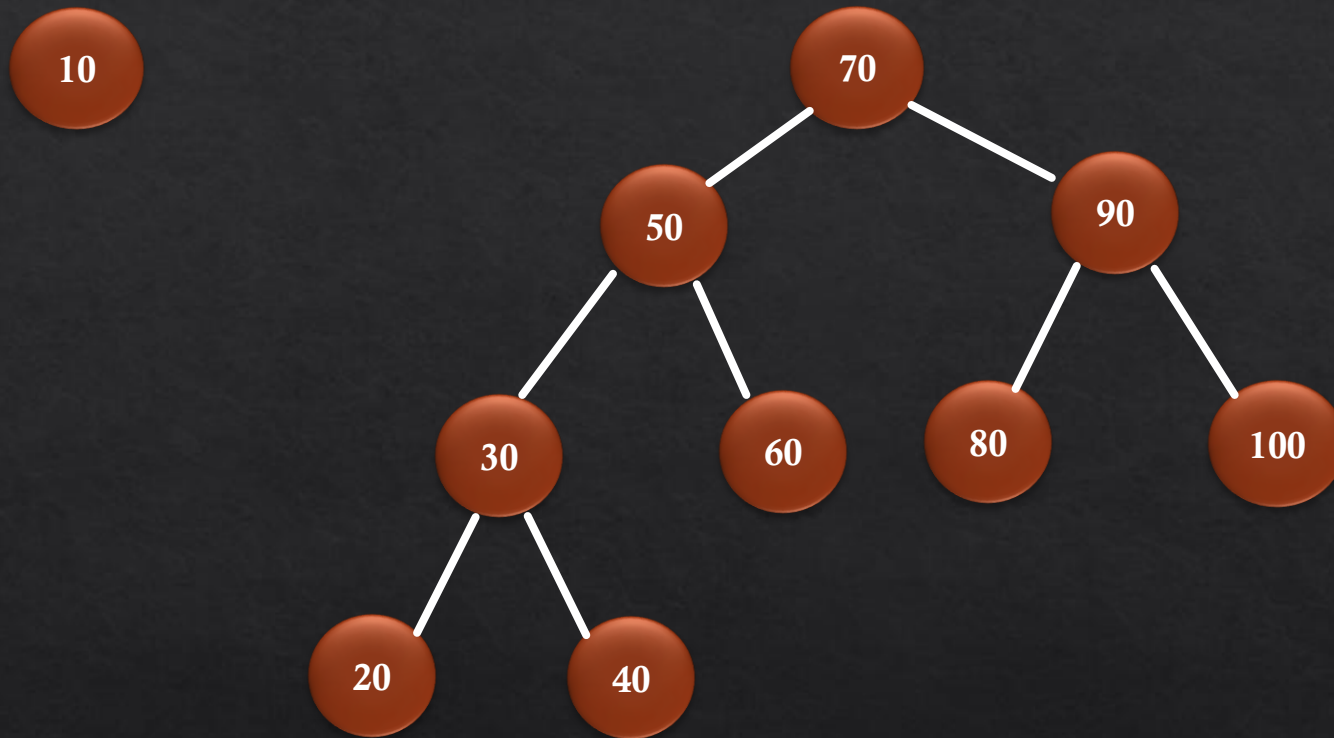
- Creation of a tree
- Insertion of a node
- Deletion of a node
- Search for a value
- Traverse all nodes
- Deletion of the tree



Creation of BST



Insertion



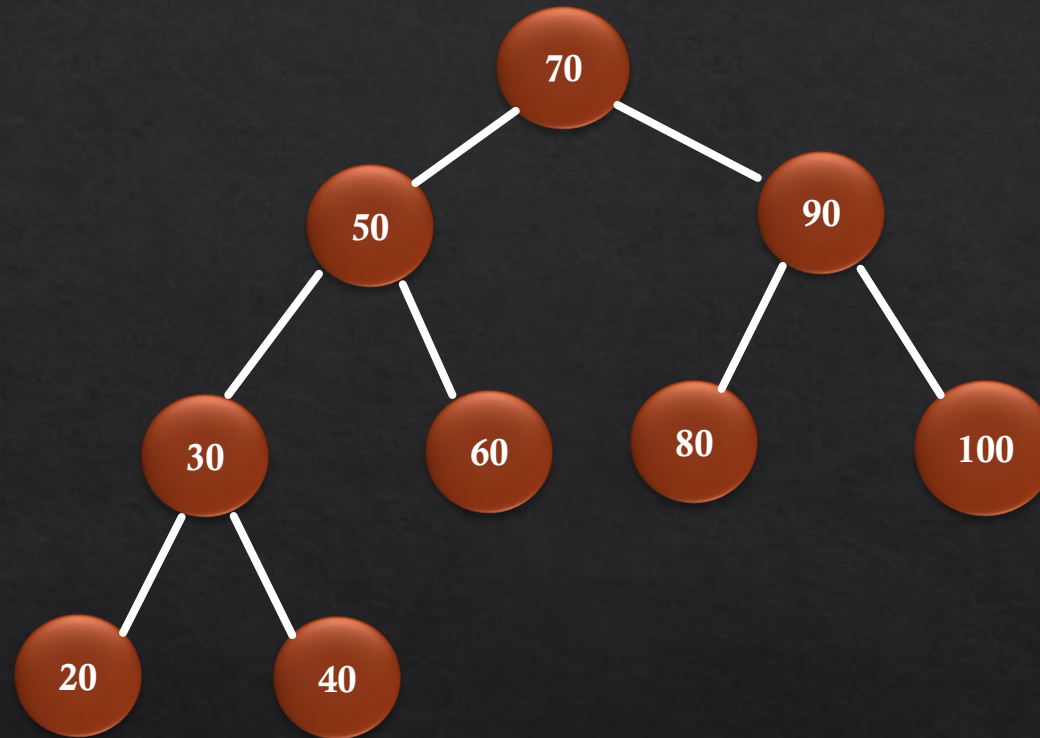
Traversal

Depth first search

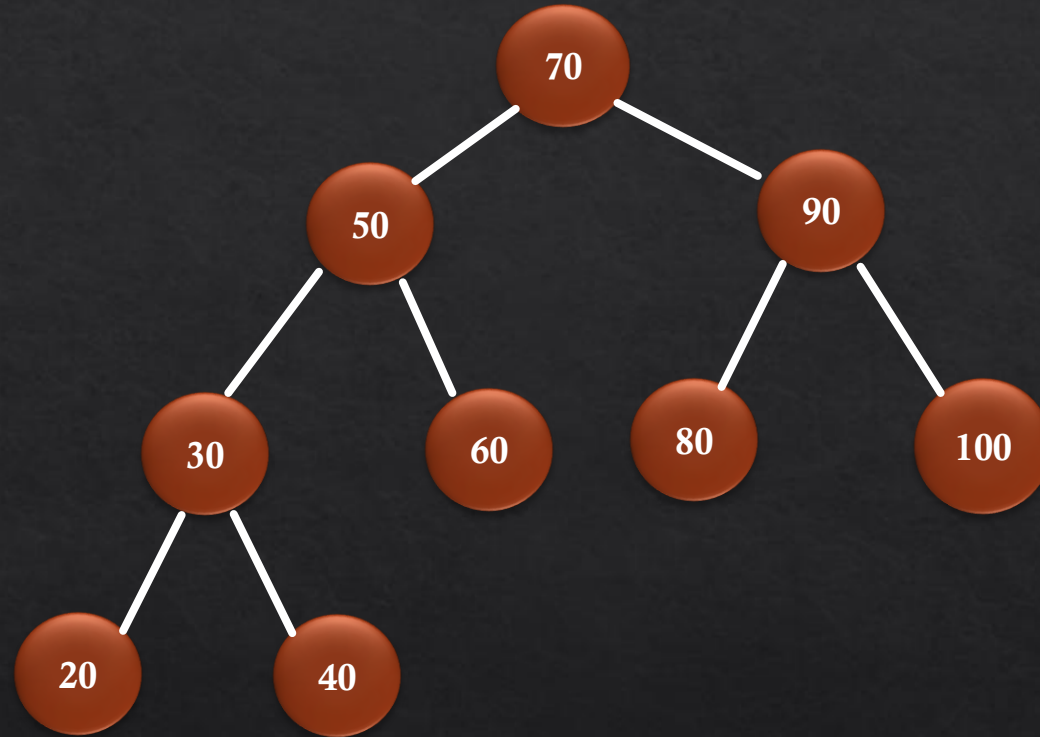
- Preorder traversal
- Inorder traversal
- Post Order traversal

Breadth first search

- Level order traversal

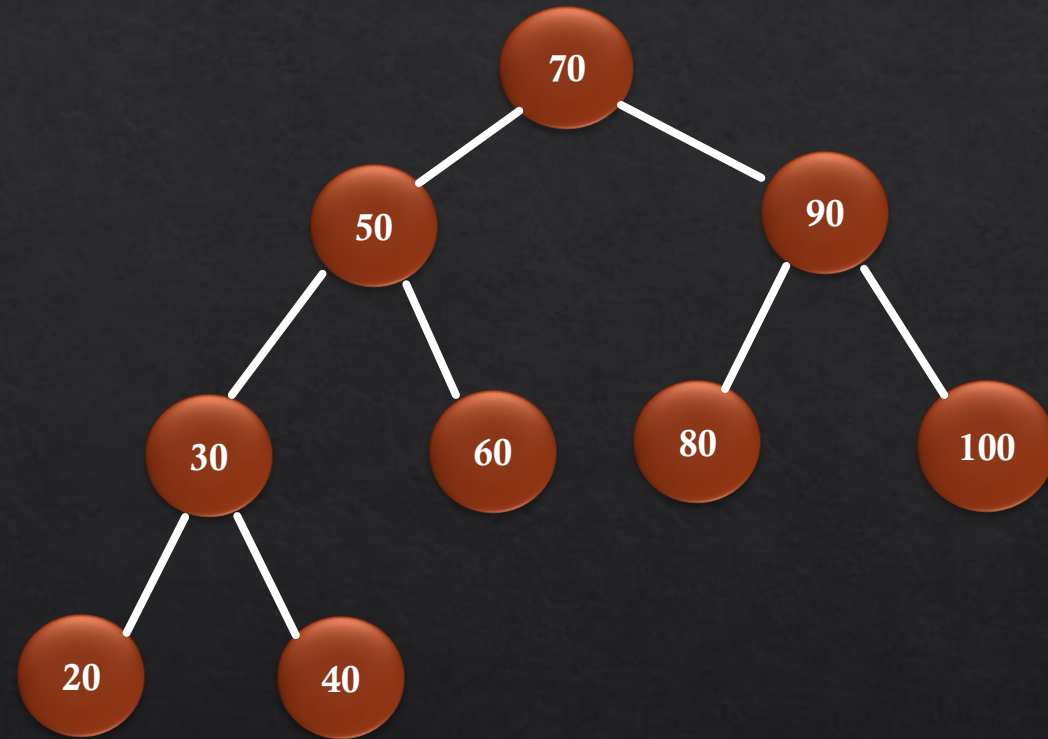


Search



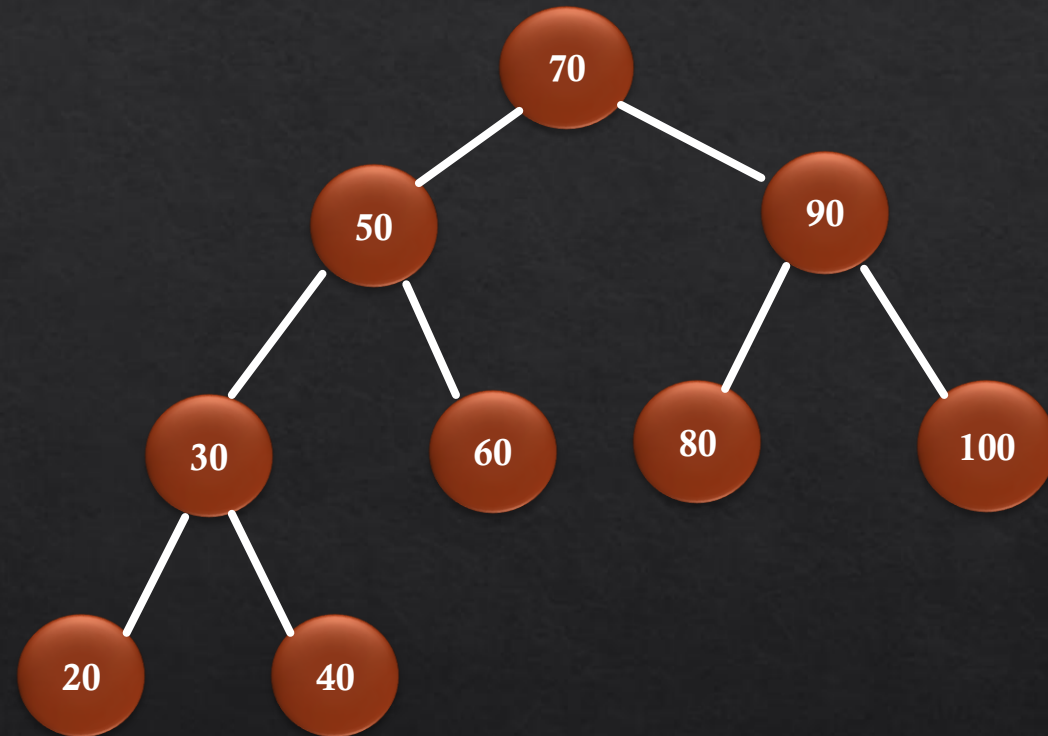
Delete a node

- Case 1: The node to be deleted is a leaf node
- Case 2: The node has one child
- Case 3: The node has two children



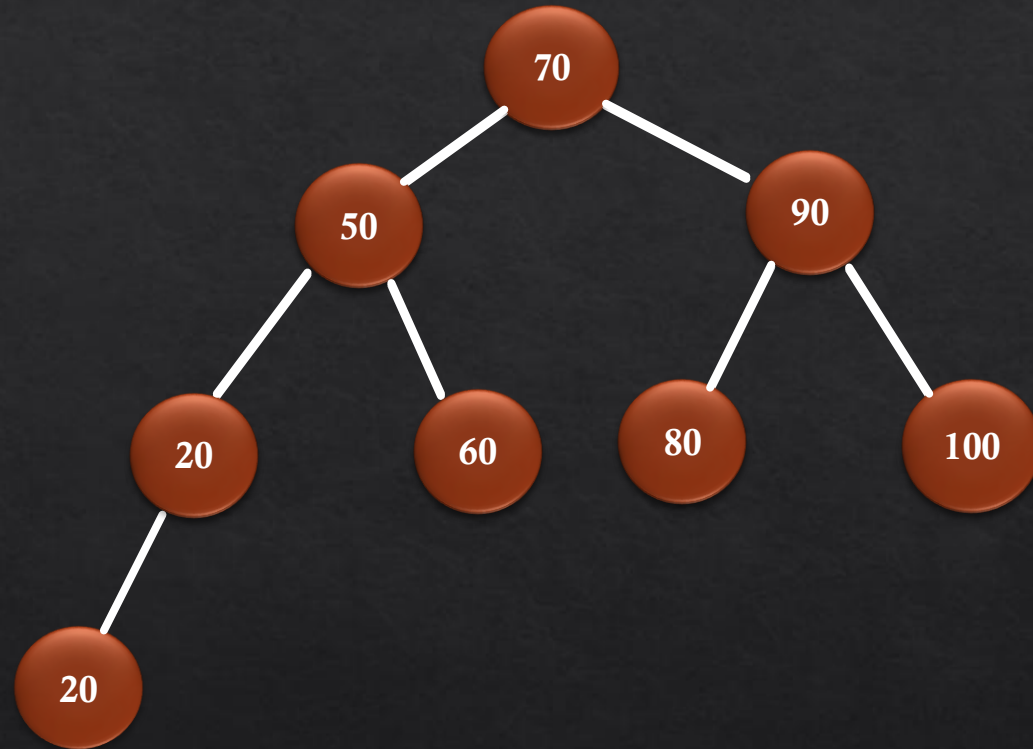
Delete a node

- Case 1: The node to be deleted is a leaf node



Delete a node

- Case 2: The node has one child

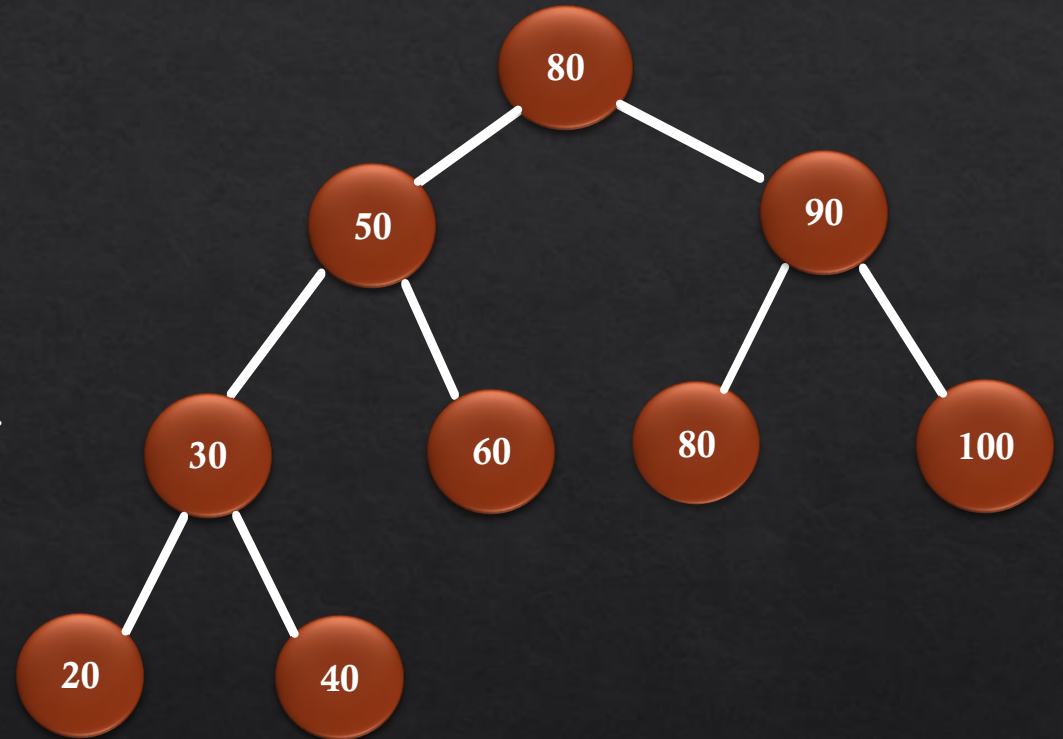


Delete a node

- Case 3: The node has two children



- Find the successor of the node.
- Successor is 80, which is the smallest node in the right subtree.

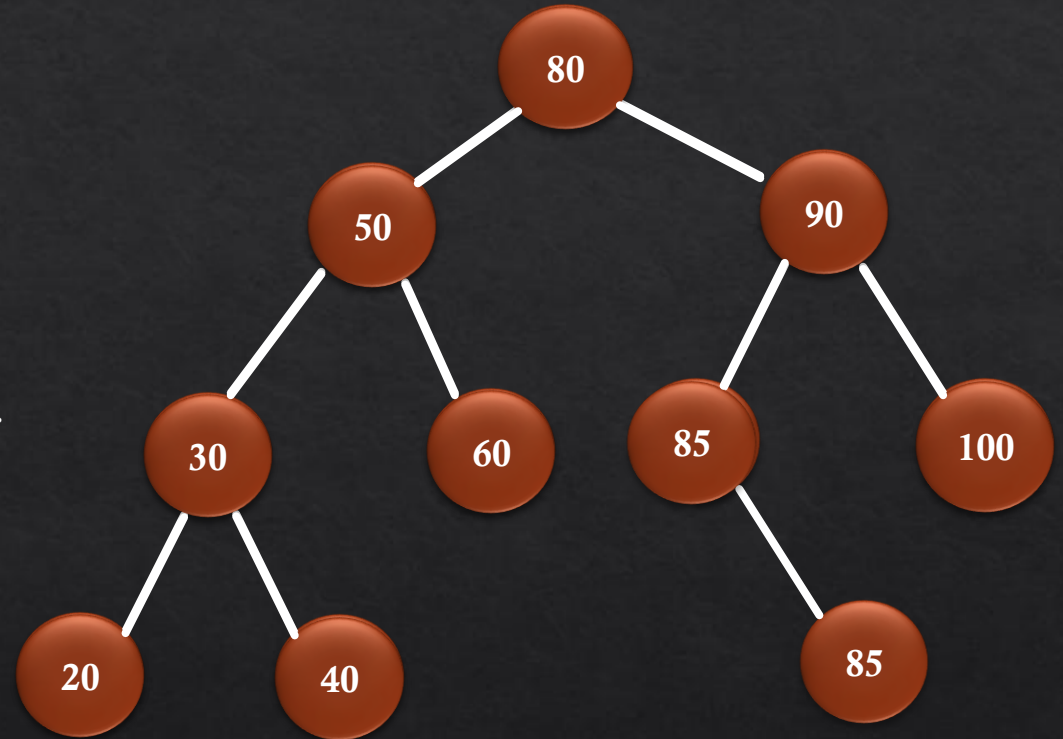


Delete a node

- Case 3: The node has two children



- Find the successor of the node.
- Successor is 80, which is the smallest node in the right subtree.



Delete entire BST

rootNode = None

rootNode.leftChild = None

rootNode.rightChild = None

